Drug Repurposing Hypothesis Assessment: Enpatoran (TLR7/8 antagonist) in Sjögren's Syndrome

Summary

The hypothesis is biologically plausible but unproven, and the case is best judged moderate-to-weak. Enpatoran is a first-in-class, selective oral Toll-like receptor 7/8 (TLR7/8) antagonist. The proposed mechanism — interrupting TLR7-driven plasmacytoid dendritic cell (pDC) type I interferon (IFN) production and B-cell activation that feed the type I IFN signature and autoantibody production — is genuinely active in Sjögren's syndrome, but the evidence places three important qualifications on it. First, the TLR7 pathway is **elevated only in the IFN-signature-positive subgroup**, roughly half of patients (PMID: 27672125). Second, **human genetics implicate the downstream IFN axis (IRF5, STAT4), not the TLR7 locus itself** (PMID: 28076899). Third, and most problematically for the repurposing case, the disabling **glandular pathology is dominated by a parallel type II IFN (IFNγ) / cytotoxic CD8 T-cell compartment that TLR7/8 blockade does not reach** (PMID: 33572487, PMID: 41162286).

There are **no clinical data for enpatoran in Sjögren's syndrome** — it has never been trialed in this disease. Its development program is in systemic/cutaneous lupus (WILLOW, NCT05162586) and COVID-19 pneumonia (NCT04448756). The most informative precedent in Sjögren's is hydroxychloroquine, an endosomal TLR7/9 inhibitor, which in the phase III JOQUER trial **lowered the IFN signature and serologic markers yet failed its primary clinical endpoint** (PMID: 25027140, PMID: 31237947). This dissociation between biomarker modulation and clinical benefit is the single most important cautionary signal for the enpatoran hypothesis.

On the supportive side, enpatoran shows target engagement and efficacy signals in mechanistically related IFN-driven lupus (PMID: 41136221, PMID: 42107375), upstream RNA/IFN-pathway targeting (RSLV-132) improved fatigue in Sjögren's (PMID: 32798283), blocking type I IFN improves Sjögren-like features in mice (PMID: 34867938), and the B-cell arm of the mechanism is clinically addressable (ianalumab; PMID: 30826774). The net verdict: enpatoran

has a credible but narrow path to benefit — most plausibly on systemic disease activity, autoantibodies and fatigue in biomarker-enriched, early-stage, IFN-positive patients, and least plausibly on restoring glandular secretion in established disease.

Key Findings

Finding 1 — The TLR7 pathway is elevated in Sjögren's, but confined to the interferon-positive subgroup

The drug's target pathway is demonstrably active in Sjögren's syndrome, but not uniformly. In a study of 115 primary Sjögren's syndrome (pSS) patients versus 42 healthy controls, endosomal TLR7 — **but not TLR9** — and its downstream signalling machinery (MyD88, RSAD2, IRF7) were upregulated specifically in IFN-positive pDCs and monocytes. Critically, the authors state: *"We found an upregulation of endosomal toll-like receptor (TLR) 7, but not TLR9, in IFN-positive (IFNpos) pDCs ($p<0.05$) and monocytes ($p=0.024$)"* and *"The upregulation of TLR7, its downstream signalling pathway, DDX58/RIG-I and IFIH1/MDA5 were confined to patients with IFN-positive pSS"* (PMID: 27672125). In the IFN-negative subgroup, TLR7 was normal and TLR9/RIG-I/MDA5 were actually *decreased*. This matters enormously for a repurposing strategy: roughly half of pSS patients would have a biologically relevant target, and the other half would not.

The pathway is also active in the affected tissue. In labial salivary glands, *"Mononuclear cells (MNCs) from pSS patients' LSGs showed TLR-7-dominant expression. B cells, plasma cells and plasmacytoid dendritic cells (pDCs) co-expressed with TLR-7"* (PMID: 30446998). This is the strongest single piece of evidence *for* the hypothesis: the drug's exact target (TLR7) is expressed in exactly the cell types (pDCs, B cells, plasma cells) the mechanism names, and in the exact tissue (salivary gland) that defines the disease.

Inference (labelled as such): The confinement of pathway activity to the IFN-positive subgroup implies that any enpatoran trial must enrich for IFN-signature-positive patients to have a reasonable chance of detecting benefit — an unselected trial would dilute the effect across patients in whom the target is not engaged.

Finding 2 — Hydroxychloroquine lowers the IFN signature but fails clinical endpoints: a direct cautionary precedent

Hydroxychloroquine (HCQ) is the closest existing analogue of enpatoran's mechanism — it inhibits endosomal TLR7/9 signalling. Its mechanism is to directly bind nucleic acids and mask the TLR-binding epitope (PMID: 24342772, PMID: 21398612), so it suppresses the same upstream node that enpatoran targets. In the phase III JOQUER randomized controlled trial (n=120), HCQ 400 mg/day versus placebo met its primary endpoint in only *"17.9% (10/56) in the hydroxychloroquine group and 17.2% (11/64) in the placebo group (odds ratio, 1.01; 95% CI, 0.37-2.78; P = .98)"* (PMID: 25027140).

The re-analysis of the same trial is the crux of the cautionary signal. HCQ *did* engage the biology — it significantly reduced IFN scores, interferon-stimulated gene expression, ESR, IgG and IgM — yet *"HCQ reduced several laboratory parameters, but failed to improve clinical response"* (PMID: 31237947). This is the central problem for the enpatoran hypothesis: in Sjögren's syndrome, lowering the type I IFN signature has already been shown, with a drug acting on essentially the same node, to be insufficient for clinical benefit. Enpatoran is a more potent and selective inhibitor, so it is not a perfect read-across, but the precedent sets a high bar: biomarker modulation is necessary but demonstrably not sufficient.

Finding 3 — Enpatoran shows target engagement and efficacy signals in related TLR7/IFN-driven diseases (CLE/SLE)

Enpatoran's clinical promise comes entirely from lupus. In a randomized placebo-controlled phase Ib study (NCT04647708, n=25), the drug was well tolerated with rapid, dose-dependent reductions in the IFN gene signature, and *"Greater reductions in Systemic Lupus Erythematosus Disease Activity Index 2000, Physician's Global Assessment and 28-joint count were observed at week 24 with enpatoran versus placebo"* (PMID: 41136221). The phase 2 WILLOW trial (NCT05162586) evaluated 25/50/100 mg twice daily over 24 weeks in cutaneous and systemic lupus cohorts; in Cohort A, enpatoran produced *"a significant, dose-dependent effect on CLASI-A score"* with adjusted mean changes from baseline up to −72 percentage points versus −44 with placebo (p=0.0002 for dose-response) (PMID: 42107375). TLR7/8 inhibition also blocks TLR8-driven neutrophil IL-8 secretion and NETosis in vitro and affected neutrophil counts in a COVID-19 trial (PMID: 40397760).

Transferability (inference): Cutaneous lupus and skin/joint lupus endpoints respond to enpatoran, and these are driven by type I IFN. Sjögren's shares the type I IFN axis, so the *systemic and serologic* components of Sjögren's are the most plausible beneficiaries. However,

the lupus efficacy is measured on skin (CLASI) and joint endpoints that have no direct equivalent in the glandular pathology that defines Sjögren's. The read-across supports benefit on systemic activity and skin/joint domains, not on sicca.

Finding 4 — Targeting upstream RNA-immune-complex/IFN drivers (RSLV-132) improved fatigue in pSS

The pathway's clinical relevance in Sjögren's itself is supported by RSLV-132, an RNase-Fc fusion that degrades the RNA immune complexes that drive TLR7 activation — i.e., it acts one step upstream of enpatoran on the same axis. In a phase II RCT (n=30; 20 RSLV-132 vs 8 placebo), *"Administration of RSLV-132 improved severe fatigue, as determined by 4 independent patient-reported measures of fatigue, in patients with primary SS"* (FACIT-F P=0.05, DSST P=0.02) by day 99, whereas placebo showed no change (PMID: 32798283). This is the most encouraging Sjögren's-specific signal that modulating the RNA/IFN axis can change a patient-relevant outcome — notably fatigue rather than glandular function.

Finding 5 — Human genetics implicate the IFN pathway (IRF5, STAT4) but not TLR7 directly

Genetic support for the mechanism is indirect. A genome-wide association study in the SICCA cohort (1,405 cases) found: *"We observed significant associations in established regions of the major histocompatibility complex (MHC), IRF5, and STAT4"* (PMID: 28076899). IRF5 and STAT4 are *downstream* of TLR7 in the IFN-production cascade, so the genetics implicate the IFN pathway broadly but do not single out the TLR7 receptor itself. The overall genetic architecture is modest: *"only a number of genes have been identified to be associated with SS, and these have only a weak or moderate effect"* (PMID: 28476436). SLE and Sjögren's share many IFN-pathway susceptibility loci (PMID: 27227345).

Inference: The absence of a top-tier TLR7 genetic signal in Sjögren's (in contrast to SLE, where TLR7 gain-of-function variants are established) weakens the human-genetics leg of the argument. The genetics support the IFN pathway as causal but place the causal weight downstream of the drug's target.

Finding 6 — Enpatoran has never been trialed in Sjögren's; its safety record is favorable but efficacy outside IFN-driven skin/joint endpoints is unproven

No enpatoran trial in Sjögren's exists in the literature. Registered programs are in SLE/CLE (WILLOW, NCT05162586), phase Ib SLE/CLE (NCT04647708), and COVID-19 pneumonia (NCT04448756). The COVID-19 phase II (n=149) is a sobering data point: *"The primary efficacy objective was not met; median time to recovery was 3.4-3.9 days across groups"* (PMID: 37873555) — another setting where target engagement did not translate into clinical efficacy on the chosen endpoint. On safety, the drug performs well: *"Serious TEAEs were numerically lower with enpatoran (50 mg 9.3%, 100 mg 2.2%) than placebo (18.4%)"* (PMID: 37873555) and in lupus *"Enpatoran remained well tolerated to week 24. Most TEAEs (96%) were mild or moderate in severity. There were no serious TEAEs"* (PMID: 41136221).

Finding 7 — The B-cell/BAFF arm of the mechanism is clinically addressable (ianalumab), but no DMARD is approved

The hypothesis names B-cell activation as a key effector. This arm is clinically tractable: in a phase II RCT (n=27), the BAFF-receptor-blocking, B-cell-depleting antibody ianalumab showed *"a similar trend showing positive therapeutic effect by ianalumab was observed across the primary clinical outcome (ESSDAI) and all secondary clinical outcomes"* (PMID: 30826774). Yet the field remains without an approved disease-modifying therapy: *"No disease-modifying therapy is currently approved for SjD, and management remains largely symptomatic"* and *"patients experience a diagnostic delay averaging roughly five to six years from symptom onset, during which irreversible glandular and systemic damage may accumulate"* (PMID: 42633339). The diagnostic-delay point directly supports the disease-stage concern (Findings 8 and 10): by the time patients are diagnosed, the glandular compartment may be beyond rescue.

Finding 8 — Compartment mismatch: glandular damage is dominated by type II IFN and cytotoxic CD8/NK cells that TLR7/8 blockade does not target

This is the strongest mechanistic argument *against* the hypothesis. The affected tissue is driven by a different arm of immunity than the one enpatoran blocks. *"Expression of both types of IFNs has also been shown in target tissues, namely in minor salivary glands where a predominance of type II IFN signature appeared to have a certain association with the development of lymphoma"* (PMID: 33572487) — type I IFN is more peripheral and linked to autoantibodies, while the gland itself runs on type II IFN (IFNγ). Single-cell and spatial transcriptomics confirm a TLR7-independent cytotoxic effector program: *"GZMK^{+}CD8^{+} T cells,*

which accumulate with disease severity, displayed a cytotoxic transcriptional program" and degranulate onto epithelial cells (PMID: 41162286). pSS pathology overall is *"largely mediated by type I and II interferon (IFN)"* (PMID: 37916875). Enpatoran blocks only the type I arm; the type II IFN / cytotoxic T-cell machinery that destroys the gland is downstream and parallel, and would be expected to continue regardless of TLR7/8 blockade.

Finding 9 — Redundant effector pathways: response to B-cell depletion tracks with T-cell cytokines (IL-17/IL-22/TNF)

Even direct B-cell depletion (rituximab, TRACTISS) only modulated glandular biology, and clinical response tracked with Th17/T-cell cytokines rather than the IFN axis. *"Rituximab treatment prevented the worsening of SG inflammation observed in the placebo arm, by inhibiting the accumulation of class-switched memory B cells within the SG"*, but *"patients classified as responders according to STAR displayed significantly higher baseline levels of C-X-C motif chemokine ligand-13 (CXCL13), interleukin (IL)-22, IL-17A, IL-17F, and tumor necrosis factor-α (TNF-α)"* (PMID: 38073013). Rituximab itself showed only modest discrimination versus placebo on ESSDAI (PMID: 34796851). The implication: pathways redundant to TLR7-IFN (Th17, TNF) drive a substantial part of the clinically relevant inflammation, so blocking TLR7 alone may leave the dominant effector arm intact.

Finding 10 — Most pSS targeted trials fail for recurring reasons; anti-CD40 (iscalimab) shows a tractable design

A review of pSS targeted therapy concludes that for most biologics *"clinical trials have found no differences versus placebo in the change from baseline in ... (ESSDAI) score and fatigue score"*, attributing failure to *"the high heterogeneous nature of pSS, irreversible exocrine glandular failure at advanced disease stages, inappropriate recruitment strategy in clinical trials, and outcome measures"* (PMID: 34731460). This is a roadmap of how enpatoran could fail even if its biology is right. By contrast, anti-CD40 iscalimab met its primary objective: *"The primary objectives were to demonstrate a dose-response relationship of iscalimab based on the change in ESSDAI from baseline to week 24 in cohort 1"* in the phase 2b TWINSS trial (PMID: 39096929), building on an earlier proof-of-concept study (PMID: 38263652). Iscalimab provides a template: an enriched active-disease population and an ESSDAI dose-response endpoint.

Finding 11 — Blocking type I IFN improves Sjögren-like features in mice, but systemic immunosuppression has been "largely inefficacious" in humans

The animal evidence is encouraging. In the MRL/lpr mouse model, anti-IFNα vaccination induced neutralising antibodies with *"type 1 IFN signature's reduction and disease features' (ocular and oral sicca syndrome, neuropathy, focus score, glandular production of BAFF) improvement, as reflected by the decrease in Murine Sjögren's Syndrome Disease Activity Index (MuSSDAI)"* (PMID: 34867938). But the human translation record is poor: *"systemic immunosuppression in SS, in contrast to other autoimmune diseases, has been largely inefficacious. Biological agents ... such as anti-TNF agents, anakinra, tocilizumab and rituximab failed to reach primary outcomes in randomized double-blind controlled trials"* (PMID: 31831255). A cathepsin S inhibitor also engaged its target but showed *"no clinically relevant improvement in ESSDAI score (primary endpoint)"* (PMID: 36864622). The mouse-to-human gap is wide in this disease.

Mechanistic Model / Interpretation

The hypothesis describes a real but **upstream and partial** node in Sjögren's pathogenesis. The following schematic places enpatoran's point of action against the full disease circuit:

```
Self-RNA / RNA immune complexes (anti-Ro/SSA)
                  |
       [ TLR7 ]  <-- ENPATORAN BLOCKS HERE (and TLR8)
                  |                      ^
         pDC activation                  |  (HCQ also acts here; failed JOQUER)
                  |                      |  (RSLV-132 degrades RNA upstream; improved fatigue
          TYPE I IFN (IFN-alpha)  ------ PERIPHERAL blood signature
                  |                      - ~50% of patients (IFN-positive subgroup)
      IRF5 / STAT4 (genetic risk)        - autoantibodies, systemic activity, fatigue
                  |
          B-cell activation  <--------- addressable (ianalumab, rituximab)
                  |
     ============================================================
     GLANDULAR DESTRUCTION (the defining, disabling pathology)
     - TYPE II IFN (IFN-gamma)  <-- NOT blocked by enpatoran
     - GZMK+ CD8 cytotoxic T cells degranulating on epithelium
     - Th17/IL-22/TNF (redundant effectors; track with response)
     - irreversible after ~5-6 yr diagnostic delay
     ============================================================
```

The diagram makes the central tension explicit. Enpatoran acts at the **top** of the cascade, on the type I IFN-generating node that dominates the **peripheral blood** and the **IFN-positive subgroup**. But the **bottom** of the cascade — the glandular destruction that causes the sicca symptoms patients actually experience — is driven by type II IFN and cytotoxic T cells that sit downstream of, and parallel to, the type I IFN arm. Blocking the top does not switch off the bottom.

This model predicts a specific, testable efficacy profile for enpatoran in Sjögren's:

Domain	Predicted enpatoran effect	Basis
Type I IFN blood signature	**Strong reduction**	Direct target engagement (F1, F3)
Autoantibodies / serology	Moderate reduction	HCQ lowered IgG/IgM (F2); B-cell upstream (F7)
Fatigue / systemic activity (ESSDAI/ESSPRI)	**Possible** benefit, IFN-positive subgroup	RSLV-132 fatigue signal (F4); lupus signals (F3)
Glandular secretion (sicca, salivary flow)	**Little to no** benefit	Type II IFN / CD8 compartment mismatch (F8); irreversibility (F7, F10)

The HCQ precedent (F2) is the empirical stress-test of the top of this model: it confirmed that reducing the type I IFN signature is achievable but, on its own, insufficient to move standard clinical endpoints. Enpatoran is more potent and selective, and the lupus data show it can move clinical endpoints *where those endpoints measure type-I-IFN-driven skin and joint disease*. Whether Sjögren's endpoints capture a comparably IFN-responsive component is the open question.

Evidence Base

PMID	Role	How it bears on the hypothesis
27672125	**Supports (qualified)**	TLR7 upregulated, but only in IFN-positive subgroup; defines the enrichment biomarker
30446998	**Supports**	TLR7 expressed in salivary-gland pDCs, B cells, plasma cells — target is in the tissue
25027140	**Challenges**	HCQ (TLR7/9 inhibitor) failed JOQUER primary endpoint
31237947	**Challenges (strongest)**	HCQ lowered IFN signature but failed clinical response — biomarker != benefit
41136221	**Supports**	Enpatoran phase Ib: target engagement + SLE efficacy signal, well tolerated
42107375	**Supports**	WILLOW phase 2: dose-dependent CLASI-A improvement in CLE/SLE
32798283	**Supports**	RSLV-132 (upstream RNA/IFN) improved fatigue in pSS
28076899	**Challenges (qualified)**	Genetics implicate IRF5/STAT4 (downstream), not TLR7
37873555	**Mixed**	COVID-19: good safety but primary efficacy not met
30826774	**Supports**	Ianalumab: B-cell arm is clinically addressable
33572487	**Challenges (strongest)**	Gland dominated by type II IFN, not type I
41162286	**Challenges**	GZMK+ CD8 cytotoxic T cells drive glandular damage — TLR7-independent
38073013	**Challenges**	Response tracks with IL-17/IL-22/TNF — redundant pathways
34731460	**Challenges**	Catalogue of recurring pSS trial-failure reasons
39096929	**Design template**	Iscalimab TWINSS: successful enriched ESSDAI dose-response design
34867938	**Supports (animal)**	Anti-IFNalpha improved Sjögren-like features incl. focus score in mice
31831255	**Challenges**	Systemic immunosuppression "largely inefficacious" in human SS

PMID	Role	How it bears on the hypothesis
36864622	**Challenges**	Cathepsin S inhibitor: target engagement, no ESSDAI benefit
42633339	**Context**	No DMARD approved; 5-6 yr diagnostic delay -> irreversibility

The evidence base is notably **asymmetric in strength**: the supportive evidence is largely *mechanistic* (target present in right cells/tissue) and *indirect* (efficacy in related diseases), while the challenging evidence is largely *clinical* (HCQ and multiple other target-engaging agents failed in actual Sjögren's trials) and *compartmental* (the disabling pathology runs on a pathway enpatoran does not block). In drug repurposing, clinical precedent in the target disease outweighs mechanistic plausibility, which tilts the overall assessment toward caution.

Limitations and Knowledge Gaps

1. **No Sjögren's-specific enpatoran data.** The entire assessment is inferential, built on read-across from lupus and from other TLR7/IFN-targeting agents. No direct efficacy, pharmacodynamic, or tissue-penetration data for enpatoran in Sjögren's exist.
2. **Tissue penetration unknown.** It is not established whether enpatoran reaches therapeutic concentrations within the inflamed salivary gland, or only suppresses the peripheral blood compartment. This gap is material because the blood (type I IFN) and gland (type II IFN) compartments are mechanistically distinct.
3. **Subgroup size and dynamics.** The IFN-positive subgroup is ~50% of patients, but the fraction in whom type I IFN is *rate-limiting* for clinical disease (as opposed to a correlated biomarker) is unknown. The HCQ data suggest the overlap may be smaller than the biomarker prevalence implies.
4. **Endpoint sensitivity.** Standard endpoints (ESSDAI, ESSPRI) may be too insensitive, too glandular-weighted, or too noisy to detect a type-I-IFN-restricted effect — a recurring cause of pSS trial failure (PMID: 34731460).
5. **Disease stage / irreversibility.** With a 5–6 year diagnostic delay, most trial-eligible patients may have irreversible glandular damage, biasing any glandular endpoint toward the null regardless of mechanism.

6. **Reliance on abstracts.** This assessment draws on abstracts and provided citation snippets rather than full texts; effect sizes and subgroup analyses could not always be independently verified.

Proposed Follow-up Experiments / Actions

1. **Biomarker-enriched phase 2 trial (highest priority).** Enroll only IFN-signature-positive pSS patients (stratify on a validated blood IFN score, per PMID: 27672125), with **early/active disease** to avoid irreversible glandular failure. Use the iscalimab TWINSS template (PMID: 39096929): ESSDAI dose-response at week 24 as the primary objective, with co-primary systemic-activity and fatigue (ESSPRI/FACIT-F) endpoints. This design most efficiently tests whether blocking the type I IFN top-of-cascade translates into clinical benefit in the subgroup where the target is engaged.
2. **Paired pharmacodynamic / tissue sub-study.** Within the trial, collect paired blood and labial-salivary-gland biopsies to quantify (a) peripheral type I IFN signature suppression, (b) intra-glandular TLR7/type-I-IFN suppression, and critically (c) whether the type II IFN / GZMK+ CD8 program (PMID: 41162286) is unaffected — directly testing the compartment-mismatch hypothesis (F8).
3. **Decouple biomarker from clinical response prospectively.** Pre-specify analysis of whether IFN-signature reduction correlates with clinical response, to confirm or refute the HCQ dissociation (PMID: 31237947) for a more potent inhibitor.
4. **Combination rationale exploration (preclinical).** Given redundant Th17/type II IFN effectors (F8, F9), model enpatoran + an effector-arm agent (e.g., anti-CD40 or JAK inhibitor) in a Sjögren's mouse model to test whether upstream + downstream blockade outperforms either alone.
5. **Fatigue/patient-reported-outcome focus.** Because the only Sjögren's-specific pathway signal to date is on fatigue (RSLV-132, PMID: 32798283), consider fatigue and systemic activity — not sicca — as the lead efficacy claim, matching the domains where the type I IFN mechanism is most plausibly rate-limiting.

Verdict

The case for enpatoran in Sjögren's syndrome is moderate-to-weak. The target pathway is real and present in the right cells and tissue, the drug works in mechanistically related lupus, blocking the pathway helps Sjögren's-like disease in mice, and the drug is safe. But three facts restrain enthusiasm: the target is engaged in only ~half of patients; human genetics and (decisively) the glandular pathology point to downstream/parallel compartments (type II IFN, cytotoxic CD8, Th17) that enpatoran does not block; and the closest clinical precedent, hydroxychloroquine, lowered the IFN signature yet failed its primary endpoint in a phase III Sjögren's trial. Enpatoran is therefore a plausible candidate whose benefit, if any, is most likely on systemic activity, autoantibodies and fatigue in biomarker-enriched, early-stage IFN-positive patients — and least likely on restoring glandular secretion. **The single most efficient test** is a biomarker-enriched (IFN-signature-positive), early-active-disease phase 2 trial using an iscalimab-style ESSDAI dose-response primary endpoint with co-primary fatigue/systemic-activity endpoints and a paired blood-plus-gland pharmacodynamic sub-study to confirm both target engagement and whether the type II IFN compartment is left untouched.

Highest-weight identifiers: - PMID: 27672125 — TLR7 pathway elevated only in the IFN-positive subgroup (defines the enrichment biomarker). - PMID: 31237947 / PMID: 25027140 — HCQ (same node) lowered IFN signature but failed clinical endpoints (strongest cautionary precedent). - PMID: 33572487 / PMID: 41162286 — glandular damage driven by type II IFN and cytotoxic CD8 T cells enpatoran does not reach (compartment mismatch). - PMID: 41136221 / PMID: 42107375 — enpatoran target engagement + efficacy and safety in related IFN-driven lupus (main supportive read-across). - PMID: 32798283 — upstream RNA/IFN targeting (RSLV-132) improved fatigue in pSS (best Sjögren's-specific pathway signal). - PMID: 39096929 — iscalimab TWINSS, the design template for a feasible enriched pSS trial. - **NCT05162586** (WILLOW) / **NCT04647708** / **NCT04448756** — enpatoran's actual trial program (SLE/CLE, COVID-19); note the complete absence of any Sjögren's trial.